

EXP NO : 5

DATE : 09/02/2026

APPLYING FILTERING AND PROJECTIONS

```
> _MONGOSH

> use studentdb
< switched to db studentdb
> db.createCollection("students")
< { ok: 1 }
> db.students.insertMany([
  { name: "Alice", age: 16, grade: "10th", marks: 85, city: "Chennai" },
  { name: "Bob", age: 17, grade: "11th", marks: 90, city: "Delhi" },
  { name: "Charlie", age: 15, grade: "10th", marks: 75, city: "Mumbai" },
  { name: "David", age: 18, grade: "12th", marks: 88, city: "Chennai" },
  { name: "Eve", age: 16, grade: "11th", marks: 92, city: "Delhi" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f60bd59ee4ad85e11932ec'),
    '1': ObjectId('69f60bd59ee4ad85e11932ed'),
    '2': ObjectId('69f60bd59ee4ad85e11932ee'),
    '3': ObjectId('69f60bd59ee4ad85e11932ef'),
    '4': ObjectId('69f60bd59ee4ad85e11932f0')
  }
}
> db.students.find({ marks: { $gt: 80 } })
< {
  _id: ObjectId('69f60bd59ee4ad85e11932ec'),
  name: 'Alice',
  age: 16,
  grade: '10th',
  marks: 85,
  city: 'Chennai'
}
{
```

>_MONGOSH

```
{
  _id: ObjectId('69f60bd59ee4ad85e11932ed'),
  name: 'Bob',
  age: 17,
  grade: '11th',
  marks: 90,
  city: 'Delhi'
}
{
  _id: ObjectId('69f60bd59ee4ad85e11932ef'),
  name: 'David',
  age: 18,
  grade: '12th',
  marks: 88,
  city: 'Chennai'
}
{
  _id: ObjectId('69f60bd59ee4ad85e11932f0'),
  name: 'Eve',
  age: 16,
  grade: '11th',
  marks: 92,
  city: 'Delhi'
}
> db.students.find([ $and: [[ age: { $gte: 16 } ], { grade: "11th" } ] ])
< {
  _id: ObjectId('69f60bd59ee4ad85e11932ed'),
  name: 'Bob',
  age: 17,
  grade: '11th',
  marks: 90,
  city: 'Delhi'
}
{
  _id: ObjectId('69f60bd59ee4ad85e11932f0'),
  name: 'Eve',
  age: 16,
  grade: '11th',
  marks: 92,
  city: 'Delhi'
}
```

```

    grade: '11th',
    marks: 92,
    city: 'Delhi'
  }
> db.students.find({ grade: { $in: ["10th", "12th"] } })
< {
  _id: ObjectId('69f60bd59ee4ad85e11932ec'),
  name: 'Alice',
  age: 16,
  grade: '10th',
  marks: 85,
  city: 'Chennai'
}
{
  _id: ObjectId('69f60bd59ee4ad85e11932ee'),
  name: 'Charlie',
  age: 15,
  grade: '10th',
  marks: 75,
  city: 'Mumbai'
}
{
  _id: ObjectId('69f60bd59ee4ad85e11932ef'),
  name: 'David',
  age: 18,
  grade: '12th',
  marks: 88,
  city: 'Chennai'
}
> db.students.find(
  { grade: "11th" },
  { name: 1, age: 1, _id: 0 }
).sort({ marks: -1 }).limit(2)
< {
  name: 'Eve',
  age: 16
}
{
  name: 'Bob',
  age: 17
}
studentdb>

```